

Summary

The authors Danieli et al. present a minimal model that solves an important reinforcement learning problem, the non-stationary K-armed bandit problem. Their model is novel in the way it learns and updates the values of each 'arm' of the K-armed bandit problems: instead of using Bayesian methods to track the optimal value, the model used two interacting neural network populations simulating neural firing rates. The authors showed that their model achieves on-par performance with standard Bayesian based RL algorithms with four different non-stationary K-armed bandit tasks.

The author's model provides a fresh way to solve non-stationary K-armed bandit problems using interacting neural networks, but may have overreached when connecting their model to prefrontal cortex (PFC) regions. More care should be given in outlining how the model is related to PFC regions and neuromodulatory systems. I recommend that the connections should be relegated to the discussion section given the lack of detail and empirical support for connecting the regions to their model and given that the paper does not model any empirical data linking their model's networks to neural activity.

Major comments:

1. Can the authors provide more descriptions of the Thompson sampling, UCB and epsilon-greedy algorithms in their introduction? For example, the term "near optimal regret bounds" is used to introduce Thompson sampling and UCB, but "regret" isn't formally defined until 2.1. In line with this, the authors might want to spend more time explaining why these models are not biologically plausible, especially since epsilon-greedy can be implemented with RL algorithms with more biological underpinnings such as TD learning (with respect to dopamine, for example). I think one particularly valuable aspect of the author's model is that it is a *minimal model* that is biologically realistic, compared to these other algorithms which itself have no biological underpinnings. It may be useful to highlight that aspect in the introduction as well.
2. The authors mentioned in their second to last paragraph of the introduction that the work aims to "bridge the gap between adaptive decision making under uncertainty and computational neuroscience." While their model have done this, there have been many, many models in computational neuroscience that address adaptive decision-making under uncertainty. These include the cited work of Wang et al. 2021, and many more literature in dopamine modeling: see Starkweather et al. 2018, Iyaga 2016 (which also models a K-armed bandit problem), Behrens et al. 2007 and Soltani and Izquierdo 2019 for a review. The introduction might be improved if the authors consider some of these existing computational models of learning under uncertainty, and how their minimal model improves upon them (in my view, their model does improve upon them given the non-stationary nature of their K-armed bandit task).
3. The authors linking the anterior cingulate cortex (ACC) and orbital-frontal cortex (OFC) to the two neural network populations of their model is certainly compelling, but the

inspiration of OFC and ACC seems to be limited to broad-strokes descriptions of the roles of these two prefrontal areas.

- a. Beyond the studies cited by the authors, the role of OFC and ACC in value learning is still under active investigation and debate (for example, see: Knudsen & Wallis 2022 for a review on OFC, see Kolling et al. 2016 for a discussion on dACC).
 - b. The authors also cited 2 papers to link their option selection process to PFC circuitry on page 6. Puzzlingly, the first paper is an PET scan study on humans looking at D2 receptors in striatum (Bäckman et al. 2011), which I presume is miscitation. The second citation is a study on unit and population dynamics of OFC and ACC in monkeys (Enel et al. 2020). The latter paper might shed light more on population dynamics of ACC and OFC neurons, but the results seem to support that a combination of dynamic and stable value representations were found in ACC and OFC (rather than what the authors here are proposing, that “choices emerge from network state sampling following a period of autonomous neural activity”).
 - c. Given that these links to ACC and OFC are fairly tenuous, I urge that the authors move these citations and paragraphs to the discussion portion of the paper. Even without the direct link to ACC and OFC, the authors still present a novel and compelling way to solve K-armed bandit tasks. That the model can perform well with its plasticity mechanism is a worthwhile contribution to the literature.
 - d. In the discussion, the authors can then link how their neuronal populations might be mapped onto ACC/OFC areas as a proposed hypothesis of how they might interact. If the authors would like to more clearly link their model to PFC areas, perhaps they could provide testable hypotheses from their model and what type of experiments might help test them.
4. In the discussion, the authors mentioned that dopamine might be a candidate for meta-plasticity for updating the value populations in neural network V . This was mentioned in passing, but the paragraph would benefit with more care in how they propose dopamine perform the meta-plasticity and the anatomical understanding of how dopamine interacts with PFC regions.
- a. For the former, it is unclear to me which hyperparameter dopamine might be meta-learning on, and how exactly this learning is happening. The authors described “the choice of modulating the afferent connections to the value population V according to a non-linear function **dependent** on the synaptic weight itself” yet later explain that “This approach can be seen as a form of meta-plasticity implemented through neuromodulation, where a region external to the network affects the synaptic connections **without altering their actual weights.**” Are the changes happening dependent on the weights or by not altering the weights at all? How are the changes dependent on the weights?
 - b. For the latter, I would like clarity on how their proposed functions fit with experimental studies looking at dopamine interacting with PFC regions. There have been studies looking at how OFC neurons can affect DA neurons (Takahashi et al. 2011) and studies that look at D1 and D2 receptors in ACC

(Schweimer & Hauber 2006, Suhara et al. 2002). In general, the discussion would be strengthened with more explicit descriptions and justifications of why the authors think dopamine would be involved.

Minor:

1. The data and code availability link was not public, and thus was not accessible.
2. In section 3 of Experiments, it is unclear the relationship between trials and rounds. Can the authors explicitly state that rounds are part of a trial to clarify how reward distributions change between trials versus between rounds? Can this also be clarified in the figure caption for **Figure 2**?
3. For the KAB-sinP model, can the authors clarify what subset of the arms are kept constant vs dynamically changing, especially given that they test models on varying #'s of K arms for Table 1?
4. The final sentence in section 3.3 seems incomplete-- it's unclear what the "non-stationary" refers to.
5. Table 1 should have the tasks labeled in the table itself, and not just in the table's caption. Table 1 should also have the best model's results bolded, rather than just the author's proposed model bolded.
6. The paper refers to their model as **Model** but also uses model as a way to describe "Model entropy" for the entropy of various models to compare. This can be confusing, and I think the authors should give a name to their model to clarify. In addition, **Figure 4b's** mention of "Entropy H" and "Model Entropy" also is confusing-- can the authors clarify that the "Model Entropy" has to do with the "selection entropy" -- and what "selection entropy" means?
7. **Figure 4b** has too many lines and it is difficult to differentiate between the reward and regret lines. Can reward and regret be plotted separately? The ylim of the plots should extend a bit below so the 0 regret and entropy options can be more clearly seen.
8. **Figure 4's** labels seem to have inconsistent capitalization-- all other figures use all lower case, but entropy, reward, and regret are all capitalized for some reason. Please make all captions consistently lower case or sentence case.

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